Milestone 2

ECE 4950

Group 18

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Executive Summary

Milestone 2 developed the sensing and concept selection foundation for the Washer Mover. The team assessed a camera-based sensing approach in MATLAB to determine the number, color, shape, and position of game objects using background subtraction, segmentation, and centroid detection. The team also translated the Washer Mover customer needs into measurable engineering requirements, generated multiple mechanical concepts, and compared the three most feasible concepts with a weighted decision matrix. The rotating game board with a stationary pickup mechanism received the highest matrix score and was selected for mockup. The work showed that the vision process can produce the board state information needed for later autonomous control while identifying the mechanical and integration issues that must be verified during prototype development.



Subsystem Design

Engineering Requirements for the Camera subsystem

Customer Requirements	Engineering Requirements
Automatically find game state	After processing starts, the sensor shall determine the game state from the acquired image without manual object labeling or manual entry of object properties.
Automatically find game state	For every processed board image, the software shall populate the gameState output with the detected object count and the position, bounding box, color, and shape of each detected object.
Find number of objects	The sensor shall report the correct number of foreground game objects in at least 95% of test images.
Find number of objects	Each visible game object shall be isolated as a separate connected foreground region so that adjacent objects are not merged into one detection in at least 95% of test images.
Find object color	The sensor shall classify the required project colors with at least 95% correct classification across the test image set.
Find object position	The sensor shall store the centroid of every detected object in pixel coordinates; measured centroid error shall be no more than 5 pixels from a manual reference.
Find object position	The sensor shall store a bounding box for every detected object so that the detected regions can be visually checked against the corresponding object in the current image.
Find object shape	The sensor shall classify the required circle, square, and triangle shapes with at least 95% correct classification.

Document Hardware

The camera subsystem uses the USB webcam provided for ECE 4950, a USB connection, a computer running MATLAB 2026a with the Image Acquisition Toolbox, and a selfie light. The course handout identifies MJPG_160x120 as the default class webcam format and uses MJPG_640x480 as the example acquisition format [1]. The 640 x 480 format is used for the resolution calculation below. Figure 1 shows the acquisition and processing data path, while the photograph documents the camera, computer, USB connection, and lighting equipment used during development.

At 640 x 480 resolution and a measured horizontal field of view of 6 in, the camera provides 640/6 = 106.7 pixels/in. One pixel therefore represents approximately 0.00938 in, or 0.238 mm. Assuming square pixels, the image density is approximately (106.7)^2 = 11,378 pixels/in^2 or about 1,764 pixels/cm^2. The 5-pixel centroid error limit defined in the camera requirements corresponds to approximately 0.0469 in, or 1.19 mm, in the imaged plane. The course handout demonstrates the same method using a 20-in field of view, which gives 32 pixels/in, or 1/32 in/pixel [1].

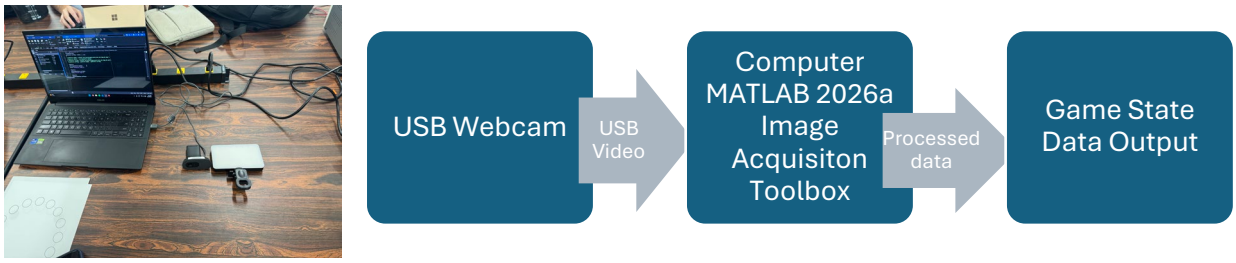


Figure 1. Camera system used for image acquisition, processing and connection diagram.

The webcam is appropriate because MATLAB can extract the required object count, color, shape, centroid, and bounding box from visible image features. Camera height, viewing angle, focus, and illumination must remain consistent during verification because changes can affect measured color and position. The selfie light provides 2700 K, 4000 K, and 6500 K settings; one setting should remain fixed during testing for repeatable background subtraction and color classification.

Document Software

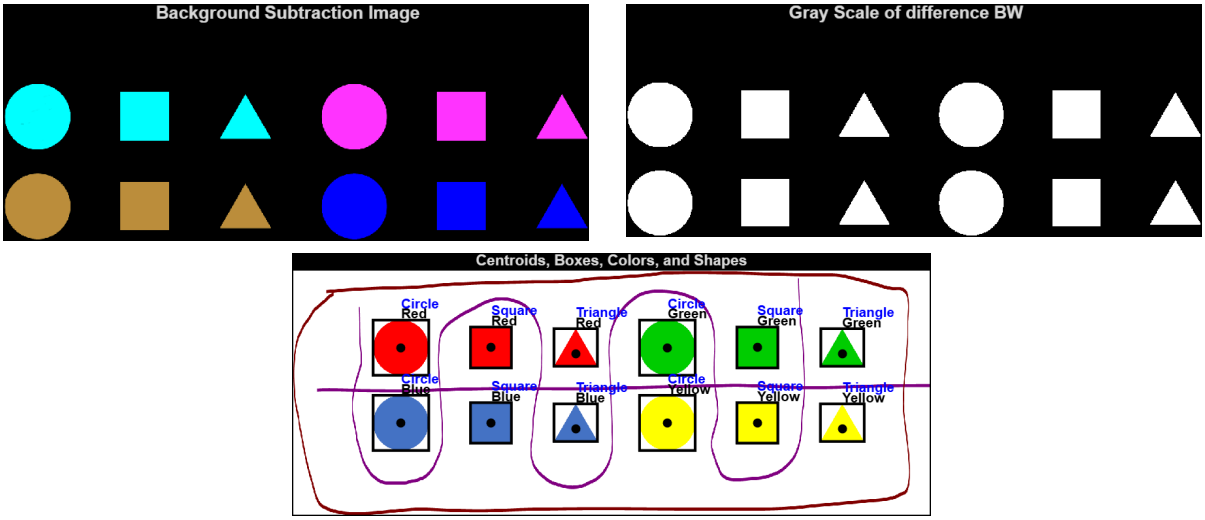


Figure 2. MATLAB image processing results showing detected colors, shapes, centroids, background subtraction, and binary foreground image.

Figure 2 shows the principal intermediate and final outputs produced during camera testing. A background image represents the board before game objects are introduced, while the current image contains the objects that must be detected. The software forms a difference image between these frames so that unchanged board pixels are suppressed, and newly introduced regions remain. The difference is converted to a binary foreground mask, and noise removal is applied before connected regions are treated as candidate objects [1]. The displayed binary image therefore provides a direct check that each physical game piece has been separated from the background before classification begins.

Once foreground regions have been isolated, the software extracts measurements from each region rather than treating the image as one undifferentiated scene. The centroid gives the object location in pixel coordinates, and the bounding box identifies the image area associated with that object. Color is read from the current color image and compared with the color references stored in colorKey, while shape is assigned from the measured properties of the segmented region. The labeled output in Figure 2 combines

these results so that object identity, location, and the quality of the segmentation can be checked in the same view.

After the foreground objects are isolated, MATLAB regionprops are used to collect geometric information for each connected region. These measurements provide the centroid and bounding box needed to associate every detection with a repeatable location on the image. The program then combines those geometric measurements with the color and shape classification results and stores them in a structure named gameState. Keeping the results in one structure prevents the later controller from having to repeat image processing and creates a defined interface between the sensing and motion portions of the project. Table 1 summarizes the fields currently stored in gameState.

The main images are also retained inside the structure because they are useful during verification. bgImg preserves the reference scene, currImg preserves the scene being analyzed, diffImg records the pixel-level difference between the two, and diffImgBw stores the binary mask used for connected component analysis. diffProp contains the region measurements returned from the segmented mask. Together, these fields allow a detection error to be traced backward from the final object label to the image and region that produced it instead of leaving only the final classification result.

The remaining fields represent the board state in a compact form, stickerNum stores the number of detected objects, stickerPos stores the centroid coordinates, stickerBox stores the corresponding bounding boxes, stickerColor stores the color label, and stickerShape stores the shape label. The entries for a detected object must remain associated across these arrays so that the controller can refer to one game piece and retrieve all of its properties. This organization is especially important when multiple objects share a color or shape because position then becomes the distinguishing property.

Table 1. gameState data

Data field / category	Purpose
bgImg	Initial image of the game board background without any washers
bgSize	Background image size [Height, Length, RGB]
currImg	Image of the game board with all washers
diffImg	Image of the bgImg – currImg to see newly changed information
diffImgBw	Black and white image of diffImg used for processing
diffProp	All regionprops of the diffImgBw
colorKey	Dictionary of color names and their respective hue values
stickerNum	Number of items detected in diffImgBw
stickerPos	Centroid location of every detected item [X, Y]
stickerBox	Bounding boxes of every detected item
stickerColor	Color of every detected item grabbed from currImg [String]
stickerShape	Shape of every detected item [String]

During continuous operation, the acquisition and analysis sequence repeats for incoming frames. Each completed pass updates gameState, providing a current description of the objects visible to the camera. In the final system, the controller can compare this detected state with the requested target pattern, determine which disk is not in its required location, and generate the corresponding motion command. The vision subsystem therefore does not directly move a disk; its responsibility is to convert the board image into consistent state information that the control software can use. Verification of the software should compare stickerNum, stickerPos, stickerColor, and stickerShape with manually established references for the same test images and should also confirm that repeated processing of an unchanged image produces stable results.

Overall System Design

Engineering Requirements for the Entire System

Table 2. Customer and engineering requirements / verification tests

Customer Requirement	Proposed Engineering Requirement	Feasible Verification Test
Autonomous	After a valid target pattern is entered and Start is activated, the system shall complete the pattern without operator intervention in at least 9 of 10 trials.	Run 10 predefined target patterns. Record every instance of manual assistance. Requirement passes if at least 9 trials need no assistance.
User-defined pattern	The interface shall allow the user to specify the desired disk state for every available well before motion begins.	Enter at least 5 different valid patterns, including sparse and full patterns. Confirm that the controller stores and executes each requested pattern.
Colored disks	The system shall manipulate every supplied disk color with at least 95 percent successful pickup and placement and without visible disk damage.	Perform 20 transfer cycles for each supplied disk color. Count successful transfers and inspect disks for damage.
Shallow wells hold a single disk	At least 95 percent of placements shall leave exactly one disk fully seated in the intended well without contacting an adjacent disk.	Perform 40 placements distributed across the board. Visually inspect each placement and count correctly seated disks.
Smooth and controlled/ Move	At least 95 percent of commanded moves shall occur without dropping the carried disk or displacing a disk that was not commanded to move.	Run 40 representative moves while observing the board. Record drops, unintended contact, and unintended disk displacement.
Quiet	Normal operation shall produce no more than 60 dBA measured approximately 1 m from the system.	Operate the system in a quiet room and record sound level during three cycles with a phone sound meter application. Use the highest measured value.
Fast/ Efficient	Average disk transfer time shall be 10 s or less from the start of pickup motion to completed placement.	Time at least 10 transfers with a stopwatch or phone video. Divide total transfer time by number of completed transfers.
Safe	A user Stop command shall halt powered motion within 1 s.	During 10 operating trials, activate Stop at different points in the motion. Use phone video or a stopwatch to verify stopping time.
User friendly	A first time user shall be able to enter a valid pattern and start a cycle within 60 s after no more than 2 min of basic instruction.	Give a new user a 2 min explanation, then time pattern entry and start. Record input errors and completion time.
Indicators	The system shall clearly communicate Ready, Running, Complete, and Fault states from a viewing distance of 1 m.	Trigger each state individually and have an observer identify the state from 1 m without touching the system.

Design Solutions

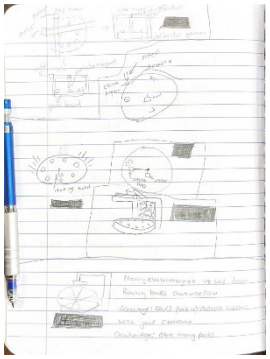


Figure 3. Team concept generation sketches

Brainstorm Concept	Brief Description	Main Strength	Main Limitation
Rotating game board, static plunger attached to electromagnet	A motor mounted to the bottom of the game board rotates the pieces into position below the elecromagnet attached to a device that goes up and down.	Simple to implement, only 3 parts.	static plunger may be hard to implement or expensive.
CoreXY motion system attached to plunger, and electromagnet	A coreXY motion system moves a device that goes up and down with an electromagnet attaced to the end.	The x, y motion system give percise X, Y control when combined with the device moving in the Z direction.	Would have a large footprint, CoreXY system is expensive, static plunger may be difficult to implement.
Electromagnet underneth gameboard attached to wiper	An electromagnet is attached underneth the gameboard and is moved via a motor attached to a rod moving the pieces.	Simple to design and only two systems to control.	Shallow wells and friction may prevent reliable transfer.
Rotating game board clockwise/ ccw, moving elecromagnet up down, multiple washers stacking	Game board is rotated via a motor, an elecromagnet attached to a device that goes up and down picks up multiple pieces in a stack much like a stack based memory seystem execpt washers.	Effiecient in picking up and placing washers in correct postions, simple to design outside of stack.	static plunger may be hard to implement or expensive, stack system would be complex and electromagnet may cause issues due to magnetic field strength.
Cartesian motion system, electromagnet	A cartesian motion system (x, y, z motion) moves an electromagnet to move pieces.	X, Y, and Z axis of control in one system.	Cartisian motion systems are large and expensive.
Motor shaking game board until pieces fall into place	A motor rotates back and forth shaking the board until the piece moves into position.	Only one part to conrtol.	Would be very hard to implement a percision shaking system and may be dangerous.

Table 3. Brainstormed concepts with strengths and limitations.

The team used brainstorming to generate several mechanical architectures capable of moving the supplied disks between wells located at a fixed radius. Figure 3 records the initial sketches, while Table 3 summarizes the broader set of concepts and the principal strength and limitation identified for each. The concepts included rotating board approaches, direct Cartesian and CoreXY positioning, a magnetic wiper located beneath the board, and a concept that uses motion of the board itself to reposition pieces. The three concepts carried forward were selected because they represented distinct ways of solving the positioning and pickup problem while remaining feasible to prototype with the available time, materials, and control hardware.

Concept 1: Rotating game board with stationary vertical pickup and electromagnet. This concept divides disk transfer into rotational positioning and vertical pickup. A motor rotates the game board until the source well is aligned with a fixed pickup station. A vertical actuator then lowers an electromagnet to the disk, energizes the magnet, raises the disk clear of the well, and holds it while the board rotates to the destination position. The actuator lowers the disk into the destination well and the electromagnet releases it. The main feature of the concept is that the pickup mechanism does not need to travel in X and Y, which reduces the moving structure above the board and turns most horizontal positioning into a single controlled rotational coordinate. Its main design challenges are accurate indexing of the board, sufficient vertical clearance, repeatable magnetic pickup, and maintaining alignment between the well center and the fixed pickup point.

Concept 2: Electromagnet beneath the board attached to a moving wiper. In this architecture, the electromagnet remains below the game board and is moved by a carriage or wiper mechanism. Magnetic attraction couples the moving magnet to the selected disk on the upper surface so that the disk follows the magnet from its source well toward the destination well. The principal advantage is that no pickup head must descend from above the playing surface, leaving the visible area relatively open and reducing the need for a separate vertical actuator. The critical feature is the magnetic coupling through the board material. The design must provide enough force to pull a disk out of a shallow well and across the surface while overcoming friction, but it must avoid disturbing nearby disks. Board thickness, well depth, surface

friction, magnet spacing, and the path followed by the wiper therefore have a direct effect on transfer reliability.

Concept 3: Motor driven board motion. This concept uses motion of the board itself to cause the pieces to move until they fall into desired locations. A motor would rotate, shake, or oscillate the board so that gravity and repeated acceleration move disks across the surface. Its main attraction is mechanical simplicity because the system can act on the board without a dedicated pickup mechanism for each disk. The limitation is selectivity. The same board motion affects every free disk, so a command intended to improve the position of one piece can disturb another piece that is already correct. Performance also depends heavily on friction, well geometry, motion amplitude, and the initial arrangement of the pieces. Other concepts, including CoreXY and Cartesian positioning systems, offered direct X and Y control and could place a pickup head over any coordinate on the board. Those architectures were not carried forward because the additional rails, motors, moving mass, footprint, and multi axis control increase implementation complexity for this prototype. The three selected concepts therefore represent a useful comparison between controlled indexing, magnetic sliding, and global board motion rather than minor variations of the same mechanism.

From an integration standpoint, Concept 1 separates the positioning problem into two controlled actions: board indexing and vertical pickup. The camera subsystem can identify the source and destination objects in image coordinates, after which the control system can map the required well location to a board angle. The rotating platform then performs the horizontal positioning while the pickup head remains mechanically fixed. This reduces the number of moving axes that must be coordinated at the same time. The main interfaces to resolve are the board drive, a rotational reference or feedback method, the vertical actuator, electromagnet power control, and the mechanical support that keeps the board and pickup station aligned throughout repeated cycles.

Concept 2 reduces the number of visible moving parts above the board and could produce a compact mechanism, but the physical interface between the magnet, board, and disk introduces uncertainty that must be resolved experimentally. Magnetic force must remain strong enough through the full board thickness and over the complete travel path. The disk must also climb out of a shallow well, cross the surface, and enter another well without sticking, rotating unpredictably, or contacting adjacent pieces. Because these effects depend on friction and geometry as well as actuator position, the mechanism would require tests of magnet spacing, surface material, well edge geometry, and repeatability before its positional accuracy could be trusted.

Concept 3 has the lowest actuator complexity because the board itself supplies the motion, but it provides the least deterministic control of an individual disk. Unlike Concepts 1 and 2, the mechanism does not establish a direct one to one relationship between an actuator position and a selected piece. A motion that helps one disk may move another away from its target, creating a control problem that becomes more difficult as the number of pieces increases. A viable version would therefore need a carefully designed board geometry and an experimentally established relationship between motion direction, amplitude, duration, friction, and the resulting piece movement. Extensive repeatability testing would be required before this concept could satisfy the autonomous final state requirement.

Concept Evaluation

The matrix uses eight criteria with weights totaling 100. Position accuracy and pickup reliability receive 20 points each because the robot must select and place the correct disk consistently. Control simplicity and manufacturability receive 15 points each because the design must be practical to build and integrate. Speed and safety receive 10 points each, while noise and cost receive 5 points each. Each concept is rated from 1 (poor) to 5 (excellent); the weighted score is the sum of each rating multiplied by its criterion weight and divided by 5.

Complex Decision Matrix									
Concept	Position Accuracy	Pickup Reliability	Control Simplicity	Manufacturability	Speed	Safety	Noise	Cost	Weighted Score
Weight	20	20	15	15	10	10	5	5	Max 100
Rotating game board, static plunger attached to electromagnet	4	4	4	4	4	4	4	3	79.0
Electromagnet underneth gameboard attached to wiper	3	3	4	4	4	4	4	4	72.0
Motor shaking game board until pieces fall into place	2	2	5	4	2	2	1	5	57.0
Score scale: 1 = poor, 2 = below average, 3 = acceptable, 4 = good, 5 = excellent. Weights sum to 100.									

Table 4. Weighted concept evaluation matrix for the three selected concepts.

The rotating game board with stationary pickup received the highest weighted score, 79/100, compared with 72/100 for the under-board wiper concept and 57/100 for the board shaking concept. It was selected because it provides the strongest overall balance of positioning accuracy, pickup reliability, control simplicity, and manufacturability. The mockup should therefore show the rotating board, the fixed pickup location, the vertical electromagnet motion, and the relationship between the drive and the well positions in sufficient detail to communicate the intended final architecture. The concept is well defined enough for prototype development and avoids the friction dependent sliding and uncontrolled piece motion of the lower scoring alternatives. Robustness still must be verified through indexing repeatability, pickup force, placement, and repeated cycle testing.

Website URL for grading sheet: https://sites.google.com/g.clemson.edu/ece4950group18/home

References

[1] Clemson University ECE 4950, "Milestone 2: Game State Sensor, Concept Generation and Website Design," course project handout, 2026.

[2] Clemson University ECE 4950, "Final Project Customer Requirements: Washer Mover," course project handout, 2026.

ECE 4950 Project 2 – Sensor and Website Rubric

Use the guidelines below to complete your report and add at the end of your report.

Group Number (and Name):_____

Score	Pts		
	5	**General Format - Professional Looking Document/Preparation (whole document)** a) Fonts, margins (11pt, times new roman, single spaced. 1" margins on all sides). b) Spelling and grammar are correct c) Layout of pictures – all figures need numbers and captions and must be referenced in the text d) Follows the page limitations below. e) References. Use IEEE reference format. f) This grading sheet is included as the final page.	**O3-SA1:1**
	5	**Page 1: Title, Group Name, Group Members, and Date** **Executive Summary** (1 concise, well-written paragraph) Provide an overview of this project. Briefly describe what you did and what you learned.	**O3-SA1:1**
		Subsystem Design	
	20	**Page 2: Engineering Requirements for the Camera subsystem** (~1 page) In the context of just the Camera-as-a-Sensor, make a two column table that contains a column for the Customer Requirements (what are the functions of the sensing system?) and the resulting Engineering Requirements. Each row should contain a specific customer requirement and the resulting engineering requirement. One customer requirement may generate multiple engineering requirements. For example, the customer will want an "accurate" system, the Engineering Requirement could be 99.5% detection success.	**O1-SA5:2**
	20	**Pages 2-3: Document Hardware** (1 page) Describe and show images of the equipment used, connection diagrams, calculation of resolution – pixels per square inch/cm on game board etc. Is the camera an appropriate sensor?	**O2-SA2:1**
	20	**Pages 3-4: Document Software** (2 pages) Using Flowcharts, state diagrams, data structures etc. describe how the software is implemented. There is no need to include the source code.	**O2-SA2:1**
	30	**Overall System Design** **Page 5: Engineering Requirements for the Entire System** (~1 page) Make a three-column table that lists the Customer Requirement in the first column and the resulting Engineering Requirement(s) in the second column. Note that a customer requirement might branch to multiple engineering requirements. The third column should list the test that will be done on the prototype to verify that the design chosen meets each requirement.	**O1-SA5:2**
		Pages 6-7: Design Solutions (~2 pages) Use a brain storming session to generate concepts. Document your top three most feasible ideas with sketches and brief descriptions while explaining the main features of each concept.	**O2-SA1:4**
		Page 8: Concept Evaluation (~1 page) Use a complex (weighted) Concept Evaluation Matrix to show how the final design was chosen from the three best ideas described previously. Include a description of the weighting factors. Make sure to use at least six of your most important criteria in the matrix and be sure to provide and include the weighting factor (i.e. importance) for each criterion.	**O2-SA1:4**

		Describe your final design choice (~1 paragraph) The robot mock up will be graded based on the following criterial: a) Does it demonstrate the proposed concept in sufficient detail? b) Is it well thought out? c) Does it look like the final construction using this concept will be robust?	
		Page 9: Grading Sheet	
	20	**General Website Format** – ***provide link here:*** https://sites.google.com/g.clemson.edu/ece4950group18/home Is the website: a) Aesthetically clean b) Complete. Does it include: 1. The Team Description? 2. Reports 1 and 2? 3. Outline of future sections? 4. Proper use of graphics? c) Follow accessibility compliance at: https://siteimprove.com/en-us/accessibility/ada-compliance-website/	**O3-SA1:1**
	10	**Demo Slideshow Presentation** (~3 slides) a) What did you do to complete this project? b) How did all the team members get involved? c) How do you believe your team dynamic is?	